

Detector-Conditioned Observability of Interference Under Coarse-Grained Measurement

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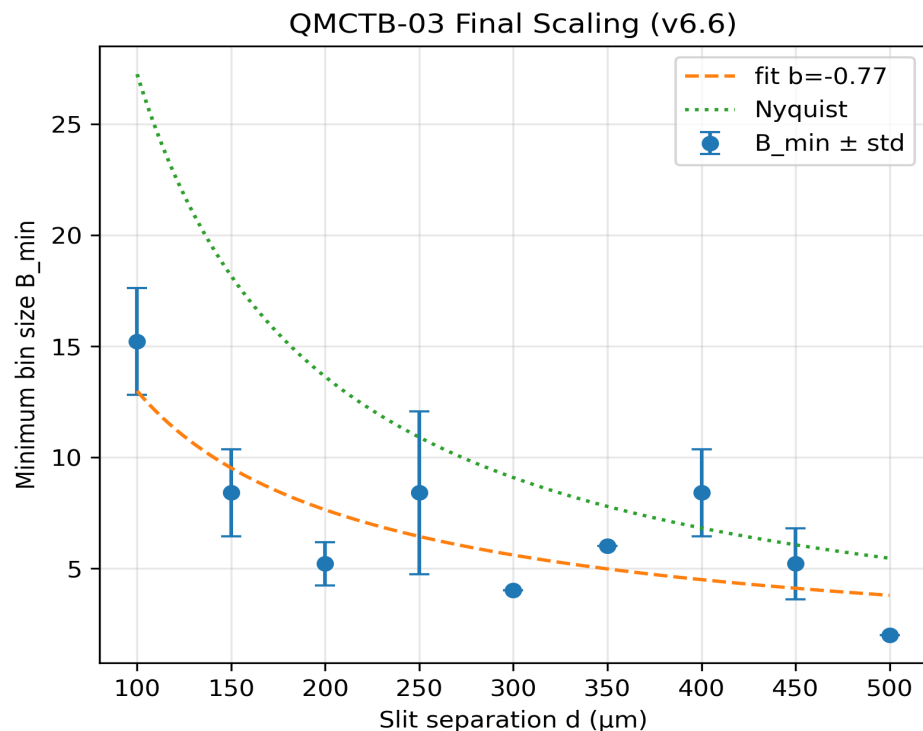
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Abstract: Interference visibility is conditioned by detector resolution and noise. We present a computational framework and observe $B_{\min} \propto d^{-0.77}$, deviating from Nyquist scaling due to noise-resolution trade-offs.

Key Result: $B_{\min} \propto d^{-0.77}$

Figure 1: Detector Resolution Scaling



Minimum detector bin size vs slit separation with fitted scaling and Nyquist comparison.

Conclusion: Interference observability is measurement-conditioned and probabilistic under noise.